Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1: Write MIPS code to perform the following integer multiplications. What is the value of the LO and HI registers?

a) **`98765` × `54321`** using the **`multu`** instruction
b) b) **`-98765` × `-54321`** using the **`mult`** instruction

```
.text
.globl main
main:
   li $t0, 98765
   li $t1, 54321
   multu $t0, $t1
   mflo $t2
   mfhi $t3
   li $t4, -98765
   li $t5, -54321
   mult $t4, $t5
   mflo $t6
   mfhi $t7
   li $v0, 10
   syscall
```

Task#2: Write MIPS code to perform the following integer divisions. What is the value of the LO and HI registers?

a) **`98765 / 54321`** using the **`divu`** instruction
b) **`-98765 / -54321`** using the **`div`** instruction

```
.text
.globl main
main:
   li $t0, 98765
   li $t1, 54321
   divu $t0, $t1
   mflo $t2
   mfhi $t3
   li $t4, -98765
   li $t5, -54321
   div $t4, $t5
   mflo $t6
   mfhi $t7
   li $v0, 10
   syscall
```

Task#3: Factorial Calculation: Using the **`mul`** instruction, write a MIPS program that computes the factorial of a number **`n`** input by the user, and display the result on the screen. Run your code and record the maximum 32-bit factorial value that can be computed without errors.

```
.data
msg: .asciiz "Enter n: "
msg_out: .asciiz "Factorial is: "
.text
.globl main
main:
    li $v0, 4
    la $a0, msg
    syscall

    li $v0, 5
    syscall
    move $t0, $v0

    li $t1, 1

factorial_loop:
    beq $t0, 0, print_fact
    mul $t1, $t1, $t0
    addi $t0, $t0, -1
    j factorial_loop

print_fact:
    li $v0, 4
    la $a0, msg_out
    syscall
    move $a0, $t1
    li $v0, 1
    syscall
    li $v0, 10
    syscall
```